Curriculum Vitae

Jialu Jacquelyn Sun

jsun488@uwo.ca | GitHub | Website

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A. RESEARCH INTERESTS

Uncertainty quantification and reliable machine learning; probability calibration, conformal prediction, and probabilistic ensembles; proper scoring rules; missing-data methods and dependence modelling for longitudinal and life-history data.

B. EDUCATION

Western University, London, ON Sep 2025 - Aug 2027 (expected)
Degree: Master of Science in Statistics
Course average: 93.0% (eight graded graduate courses)
Selected coursework: Advanced Probability Theory; Topics in Statistical Inference; Generalized Linear Models; Regression; Statistical Computing; Advanced Data Analysis; Principles of Biostatistics (first in class).

University of Waterloo, Waterloo, ON 2019 - 2025
Degree: Bachelor of Mathematics, Honours Statistics, Co-operative Program; Minor in Cognitive Science
Awarded with Distinction Jun 2025
Final two-year STAT average: 87.50% (12 courses)
Selected coursework: Statistical Learning - Advanced Regression (89%); Generalized Linear Models and their Applications (87%); Statistical Methods for Life History Analysis (96%; longitudinal and survival analysis); Experimental Design (95%); Sampling Theory and Practice (95%); Algorithmic Problem Solving (97%); Computational Neuroscience (94%).

C. RESEARCH EXPERIENCE

Uncertainty and Calibration in Train-Free Probabilistic Ensembles Apr 2026 - Present
Master's Thesis | Western University | Supervisors: Hyukjun Gweon, Hao Yu, and Reginald Kulperger
- Develop train-free ensembles that select or dynamically weight calibrated forecasts by distance from the class prior.
- Analyse Brier risk using proper scoring rule decomposition to study forecast spread and calibration alignment.
- Built logit-additive simulations with calibrated marginal forecasts; P-extreme lowered mean expected Brier score from 0.1753 (equal averaging) to 0.1585 and outperformed the better base in 87.8% of matrices.

Conformal Prediction for Parkinsonian Classification Feb 2026 - Present
- Modelled three-class outcomes for 3,434 Canadian Open Parkinson Network participants using an ensemble of CatBoost and two-stage hierarchical classifiers with Dirichlet calibration.
- Evaluated class-conditional conformal sets across 21 model and missingness scenarios with three repeats of five-fold outer validation and participant-cluster bootstrap intervals; assessed log loss, Brier score, coverage, and set size.
- Assessed coverage-efficiency trade-offs: primary conformal sets retained the recorded diagnosis in 93.35% of evaluations (mean size 1.846); scenario unions raised inclusion to approximately 99% (mean size 2.289).

Undergraduate Research Assistant Sep 2023 - Dec 2023
Cognition and Natural Behavior Laboratory | University of Waterloo, Waterloo, ON
- Recommended experimental design refinements and contributed to meta-analysis methodology discussions.
- Reviewed cognitive science literature weekly to inform research discussions.

D. MANUSCRIPTS AND CONFERENCE CONTRIBUTIONS

D1. Preprints

1. Bi, J., **Sun, J.**, Dou, H., Yuan, J., Xian, C., & Espin-Garcia, O., for the Canadian Open Parkinson Network. (2026). Uncertainty aware prediction from incomplete clinical records: a conformal prediction study of Parkinsonian classification using C-OPN data. *arXiv preprint.* arXiv:2611.05812.

D2. Manuscripts in Preparation

1. **Sun, J.**, Gweon, H., Yu, H., & Kulperger, R. (2026). A train-free local selection rule for combining probability forecasts. *Manuscript in preparation.*

D3. Conference Abstract and Team Poster Jun 2026

1. Bi, J., **Sun, J.**, Dou, H., & Yuan, J. (2026). Class imbalance aware ensemble learning for clinical differentiation of Parkinson disease and atypical parkinsonism. *Statistical Society of Canada Annual Meeting, Case Study #1, Hamilton, Canada. Accepted abstract and team poster presentation.*

E. SELECTED MACHINE LEARNING PROJECTS

Mixed-Effects Trees and Random Forests for Life Expectancy Prediction | R

- Independently implemented mixed-effects tree and random forest methods in R from the research literature, adapting models for clustered data to longitudinal prediction.
- Benchmarked predictive performance against a multiple linear regression baseline using cross-validation.

YouTube Shorts Virality Analysis | Python

- Built a YouTube Data API pipeline for metadata collection and exploratory analysis; designed sampling strategies to reduce selection bias toward popular videos.

F. TEACHING EXPERIENCE

Western University, London, ON

Teaching Assistant | Regression (SS 3859A/9859) Sep 2026 - Dec 2026

- Hold office hours, grade assessments, and facilitate review sessions.

Teaching Assistant | Statistical Modelling I (MDA 9159) Sep 2026 - Dec 2026

- Lead weekly tutorials, prepare tutorial materials, grade assessments, and hold office hours.

Head Teaching Assistant | Data Science Concepts (DS 1000) Sep 2025 - Apr 2026

- Led weekly Python lab tutorials and helped develop course materials; coordinated the teaching assistant team.
- Held office hours and graded assessments.

G. WORK EXPERIENCE

Data Scientist | Omnium Sep 2024 - Dec 2024

- Built and validated regularized sales prediction and ARIMAX forecasting models using three weeks of data following price changes.
- Constructed confidence intervals for pricing metrics to support decisions with quantified uncertainty.

Statistician | Statistics Canada Jan 2024 - Apr 2024

- Integrated large datasets in SQL and used logistic regression with feature selection to model survey response.
- Built decision tree classifiers in scikit-learn, addressed class imbalance with SMOTE, and tuned and evaluated models using grid search and cross-validation.
- Recommended collection strategies including group prioritization; response rates increased from 27% to 44% in a subsequent survey.

Data Analyst | Maple Leaf Foods Jan 2023 - Apr 2023

- Used linear regression in Python to identify a previously unrecognized predictor of equipment failure; led two co-op students in classifying 800+ forms across nine factories.

Additional experience:

Software Test Co-op | BlackBerry SecuSmart Jan 2022 - Apr 2022
Business Direction Intern | Rally Assets Jan 2021 - Apr 2021

H. HONOURS AND AWARDS

- University of Waterloo President's Scholarship (2020).
- Dean's Honours List, University of Waterloo (Winter 2020; Spring 2020).
- Term Distinction, University of Waterloo (Fall 2022; Spring and Fall 2023; Spring 2024; Winter 2025).

I. TECHNICAL SKILLS

Programming and tools: Python (pandas, NumPy, scikit-learn, Matplotlib, Seaborn), R, SQL, SAS, Git
Uncertainty and evaluation: Dirichlet calibration, class-conditional conformal prediction, proper scoring rule decomposition, risk-coverage analysis, nested cross-validation, participant-cluster bootstrapping
Modelling: CatBoost, hierarchical classification, tree ensembles, elastic-net regression, mixed-effects models, time-series forecasting

J. HACKATHON EXPERIENCE

CxC Datathon 2025 | 3rd Place Mar 2025

Predictive Modelling and Analytics for Canadian Startup Investment | Python

- Developed predictive models and supporting Python code to analyse Canadian startup investment data.
- Analysed funding trends, sector growth, and investor behaviour to inform recommendations for startups and investors.